IS DATA VALUE PATH-ORDERED OR SET-ORDERED? A CARRIER-DEPENDENT SAME-SAMPLE TEST UNDER WSD SCHEDULES

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ABSTRACT

Data attribution is computed from a selected set although optimization follows a schedule. We distinguish *set-ordered* from *path-ordered* marginal value by measuring each target at multiple entry positions within the same seed. On two frozen synthetic logistic carriers the paired test returns *carrier-dependent* readings: the sine carrier shows a negative paired departure ($g = -0.186$, CI$[-0.258, -0.115]$) that *reverses*, rather than reproduces, the cross-sectional sign (negative g: early entry *lowers* value; a negative τ: early-scheduled targets rank *higher*; Section 4.2), while the linear carrier is ambiguous ($g = +0.010$, CI$[-0.112, +0.132]$). On a pure set-ordered ridge where value is order-invariant by construction, the *identical* paired estimator carries a non-negative bias ($g = +0.102$, CI$[+0.028, +0.176]$ on the canonical allocation, positive on all four allocations, minimum CI lower -0.066). Because a non-negative bias cannot produce a negative departure, the calibration is *conservative* with respect to the sine headline. The canonical allocation literally fires the sealed binary contract (CI excludes 0), but its consequence clause is two-sided (sign-blind) and the measured bias is opposite in sign to the headline, so it does not falsify the negative sine reading (a specification erratum, not a rewrite of the frozen criterion). Value-conditioned routing reproduces the cross-sectional $B2$ negative τ on a set-ordered world, so $B2$ is a reference-conditioned stress test; identification remains undecided on this zero-GPU bed.

1 INTRODUCTION

Which training samples matter is often treated as a property of a selected set. Datamodels predict model outputs from training-set membership (Ilyas et al., 2022), while TRAK provides scalable set-indexed model-behavior attribution (Park et al., 2023). Neither representation retains the path by which samples entered an optimization trajectory. This abstraction is useful, but it leaves open a prior measurement question: is per-sample marginal value actually invariant to entry order?

The question matters whenever data are introduced by a curriculum, sampler, or staged acquisition rule. If value is set-ordered, an order-invariant attribution is an adequate description of sample ranking. If value is path-ordered, the same sample set can induce different marginal-value relations under different entry schedules. Distinguishing these possibilities requires more than observing a nonzero correlation: the experiment must specify the entry construction, use the seed rather than the item as its inference unit, enforce a minimum effect size, and *measure* the detector's behavior on a known set-ordered case (Section 4.3) rather than assume it is quiet.

We introduce a frozen Construction-Sensitivity Test on a deterministic logistic-regression test bed. Figure 1 emphasizes the controlled comparison (schematic; all quantitative evidence comes from the frozen runs and the data-derived Figure 2): the same sample identities and held-out readout are used across three entry rules, each evaluated through both a path-sensitive leave-one-out value and an order-invariant set comparator.

Contributions. First, we give a same-sample measurement of per-target entry-position sensitivity, so a cross-sectional selection signal can no longer masquerade as a path effect. Second, a quantified calibration: the sine carrier shows a negative paired departure ($g = -0.186$, CI$[-0.258, -0.115]$) that *reverses* the cross-sectional sign (Section 4.2), the linear carrier is ambiguous ($g = +0.010$, CI$[-0.112, +0.132]$), and a set-ordered ridge counter-arm shows the *identical* paired estimator carries a non-negative bias ($g = +0.102$, CI$[+0.028, +0.176]$ on the canonical allocation, positive on all four allocations, minimum CI lower -0.066), so that bias cannot explain—and is conservative with respect to—the negative sine departure. Third, value-conditioned routing on a set-ordered world reproduces the cross-sectional $B2$ negative τ, making $B2$ a stress test, not a decisive effect; a discriminating-bed certificate and decoupled positive control bind this reading. We claim only a controlled measurement on two synthetic carriers.

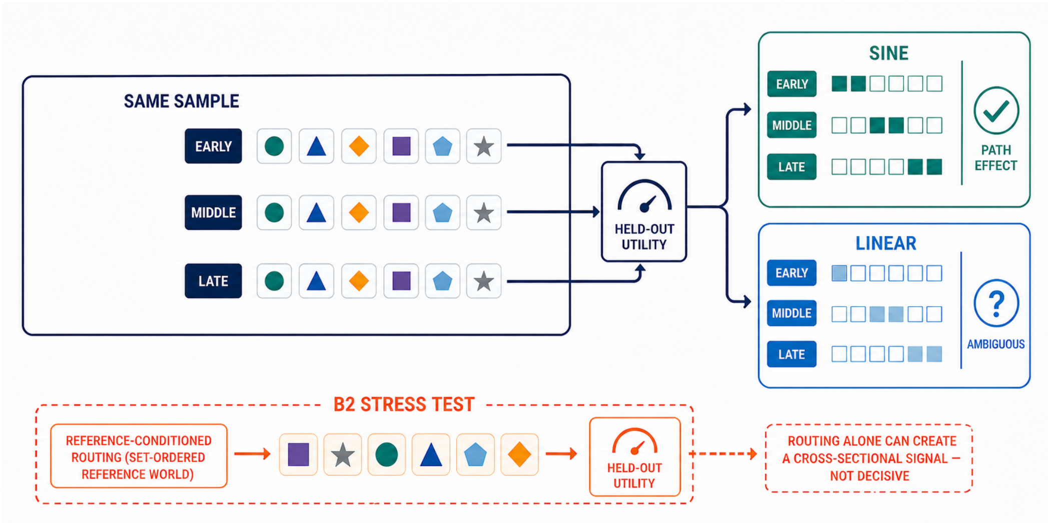

Figure 1: Carrier-dependent measurement (qualitative schematic). Each target is measured at early, middle, and late entry while its identity and held-out readout are fixed. The in-panel badges are the paired arm's frozen four-state labels (Table 5); the set-ordered counter-arm of Section 4.3 shows the same estimator's bias is non-negative, which is conservative with respect to the negative sine departure. The separate $B2$ lane is the reference-conditioned stress test: routing alone can create a cross-sectional signal in a set-ordered world, so that lane is not decisive. The audited raster contains no experimental values.

2 Problem and Estimands

For seed k, let D_k be the clean training set, G_k a fixed target set, and H_k a fixed held-out split. An entry construction c assigns each target $z \in G_k$ a schedule location $s_c(z)$. With $U(w; H_k)$ denoting held-out negative cross-entropy (higher is better), the empirical path value is

$$V_{c,k}(z) = U\left(w_{c,k}^{+G}; H_k\right) - U\left(w_{c,k}^{-z}; H_k\right), \tag{1}$$

where both fits use the same WSD budget and differ only by leaving out target z. This is a marginal utility, not the raw loss or a per-item independent replicate.

The seed-level path statistic is

$$\tau_{c,k} = \mathrm{KendallTau}\left(\{s_c(z)\}_{z \in G_k}, \{V_{c,k}(z)\}_{z \in G_k}\right). \tag{2}$$

Negative τ means that targets scheduled earlier have larger held-out marginal value. The estimand is $\widehat{\tau}_c$, the mean of Equation 2 over seeds. Frozen 95% Student-t confidence intervals are formed across the seed-level coefficients; a two-sided Wilcoxon signed-rank result is retained as a corroborating significance check for the per-sample constructions. The minimum meaningful magnitude is $\tau_a = 0.10$.

The reporting label is deliberately two-part. A cell is *null* when its interval includes zero, *sig-sub-threshold* when it excludes zero but does not lie strictly beyond $\pm\tau_a$, and *sig-supra-threshold* when the entire interval lies beyond the threshold in the effect direction; *reverse* is reserved for a significant sign reversal. Separately, `exceeds_threshold` records the minimum-effect decision. This 4-state presentation labels the outcome of the pre-registered 3-state decisive test without conflating evidence against zero with an effect large enough to trigger the headline claim.

Set-valued comparator. On the same $D_k \cup G_k$ and H_k, the order-invariant arm fits a regularized logistic model to the full set and evaluates the inverse-Hessian leave-one-out approximation

$$\widehat{V}_{\mathrm{set},k}(z) = -\nabla_w U(w_k^*; H_k)^\top H_k^{-1} g_k(z), \tag{3}$$

where H_k is the regularized training Hessian and $g_k(z)$ is the target-sample loss gradient. This is the exact operational GLM reduction used here as the nearest set-valued TRAK/datamodel comparison; it is not an implementation of their large-scale pipelines.

3 Frozen experimental design

The experiment uses two 24-dimensional synthetic logistic carriers. The linear carrier contains a clean class-separated pool and conflicting near-duplicates with flipped labels. The second carrier, retained under its frozen artifact key `sine`, uses a shifted Gaussian pool with a fixed linear label boundary; its targets negate the first feature and flip the label. We therefore use "sine" only as the registered carrier name, not as a claim of broad nonlinear coverage. The fixed configuration is summarized in Table 1.

Component	Frozen setting
Carriers	linear; registered `sine` carrier
Dimension / clean pool	$d = 24$ / 400 examples per seed
Split / targets	320 clean train; 80 held out; 40 conflicting targets
Optimizer	full-batch logistic GD, ℓ_2 coefficient 10^{-3}
WSD budget	$T = 300$; 10% warmup, 80% stable, 10% decay
Entry locations	$0.15T$, $0.50T$, $0.85T$ (or rank-linear positions for $B1$)
Inference	16 seeds; per-seed Kendall τ; 95% Student-t CI
Magnitude gate	$\tau_a = 0.10$
Negative control	20 seeded trials; Bonferroni/FWER-controlled detector

Table 1: Frozen experimental contract. Items within a seed are never treated as independent inference units. All headline intervals are computed over seed-level coefficients.

3.1 Entry constructions

Construction A inserts the complete target block from one of the three schedule locations onward and computes leave-one-out value within that block. Construction $B1$ assigns each target a distinct single injection step, linearly spread from $0.15T$ to $0.85T$. Construction $B2$ ranks targets by the fixed set-valued reference score, partitions that order into thirds, and injects the high-, middle-, and low-ranked regions at the early, middle, and late locations. Thus, $B1$ genuinely changes per-sample position, while $B2$ additionally imposes a region-lumped relation between reference value and entry time. The comparison identifies the entry rule, not the binary attribute "position was manipulated."

Worked example of Construction A (for Eq. (1)). For the 40 targets G_k and insertion fraction $t \in \{0.15T, 0.50T, 0.85T\}$, $s_A(z) = 0$ for targets outside the chosen block and $s_A(z) = \lfloor tT \rfloor$ for targets in the block, so the block is injected only at step $\lfloor tT \rfloor$ (all other targets present from step 0). The leave-one-out value is then $V_{A,k}(z) = U(w^{+G}) - U(w^{-z})$, where both fits use the same WSD budget and differ only by whether z is in the block—exactly the general definition of Eq. (1).

3.2 Controls, validity, and frozen decision

The clean negative control uses a strictly convex quadratic ridge substrate with additive, schedule-invariant influence. Independent random position labels are tested with a per-seed critical value Bonferroni-adjusted over the 20 trials; the frozen acceptance condition is no fires. The decisive result is then read per cell using both the 4-state label and `exceeds_threshold`. A global construction-dependent verdict requires that the threshold-crossing pattern vary by construction or carrier; it does not license a universal mechanism statement.

The code, specifications, and outputs were frozen before this writing pass. In particular, the current paper does not regenerate or tune the six cells, choose seeds after observing effects, or replace the seed-level uncertainty with item-level pseudo-replication. The provenance manifest and hashes are listed in Appendix B.

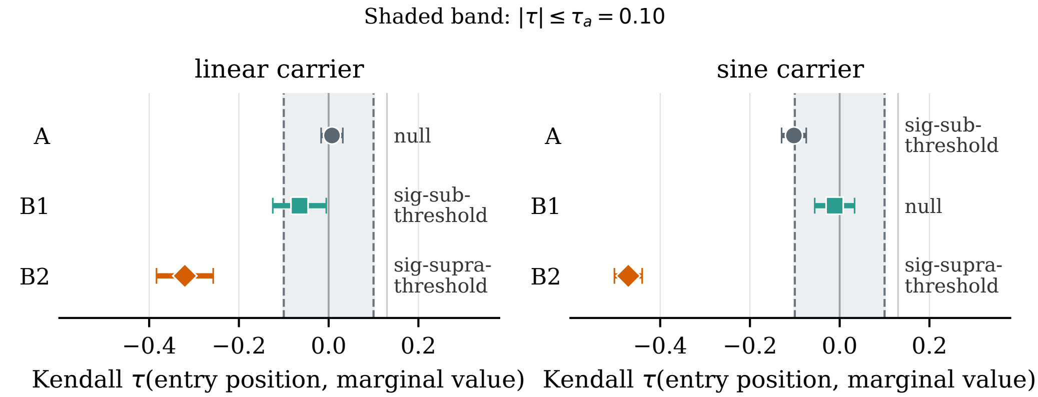

Figure 2: Construction sensitivity from the frozen six-cell artifact (16 seeds per cell). *Takeaway:* only the two $B2$ cells lie wholly beyond the negative magnitude boundary $-\tau_a$; A and $B1$ stay inside it on both carriers. Points are mean per-seed Kendall τ; bars are the frozen 95% Student-t intervals. The shaded region marks $|\tau| \leq \tau_a = 0.10$. This figure is generated deterministically from the SHA-pinned R128 JSON; exact values and the independent `exceeds_threshold` decisions appear in Table 2.

Carrier	Entry construction	τ	95% CI	4-state label	`exceeds_threshold`
linear	A: time-fraction block	$+0.007$	$[-0.017, +0.032]$	**null**	`false`
linear	$B1$: rank-linear per-sample	-0.065	$[-0.125, -0.005]$	sig-**sub**-threshold	`false`
linear	$B2$: region-lumped per-sample	-0.321	$[-0.384, -0.257]$	sig-**supra**-threshold	`true`
sine	A: time-fraction block	-0.102	$[-0.130, -0.075]$	sig-**sub**-threshold	`false`
sine	$B1$: rank-linear per-sample	-0.011	$[-0.056, +0.033]$	**null**	`false`
sine	$B2$: region-lumped per-sample	-0.471	$[-0.502, -0.440]$	sig-**supra**-threshold	`true`

Table 2: Frozen six-cell result (16 seeds per cell). The 4-state label and `exceeds_threshold` are both reported. Only the $B2$ cells cross the pre-registered magnitude threshold, and both are negative. The intervals are per-cell and unadjusted; because the six cells are not independent, we do *not* read them as a single family of six tests, and the cross-cell interval overlaps discussed below are declarations of interval overlap, not additional hypothesis tests (see Section 4 multiple-comparison note). The NEG-NC detector is SILENT, with 0/20 fires under FWER control.

4 Construction Sensitivity

Figure 2 gives the data-derived overview, and Table 2 reports the exact frozen values together with both required labels. Only the two $B2$ cells lie wholly beyond the negative magnitude boundary. Statistical significance alone would give the wrong headline: A on sine and $B1$ on linear exclude zero but remain below the minimum-effect threshold.

Construction-specific finding (cross-sectional). Cross-sectionally, $B2$ crosses $\tau_a = 0.10$ on both carriers, with CIs below $-\tau_a$; A and $B1$ do not. Routing can manufacture this pattern (Section 4.4), so the paired sine departure and ambiguous linear reading remain subject to the counter-arm caveat (Sections 4.2–4.3).

4.1 Construction-axis separation

A threshold label could differ merely because two noisy intervals straddle a boundary. The frozen construction-axis analysis therefore compares the intervals themselves. Table 3 shows that $B2$ is separated from both alternatives on each carrier. The A–$B1$ contrast is mixed: the linear intervals overlap, whereas the sine intervals are separated. This yields the pre-registered CONSTRUCTION-DEPENDENT-SEPARABLE reading.

Throughout, the cell and construction-axis intervals are *seed-level, within-cell* reads treated as *declarations of interval overlap*, not as an unadjusted multiple-testing family (cells share seeds, so their intervals are correlated). Inferential cross-construction comparisons are instead *within seed* via the paired same-sample identification of Section 4.2; claims that a contrast "excludes zero" or "is separated" come from the paired seed-level difference, with multiplicity declared, rather than from reading uncorrected per-cell intervals as independent tests.

Because the six cells share seeds, the *between-cell* interval comparison above is conservative (the two trajectories in the same carrier are positively correlated). We therefore supplement it with the *same-seed paired difference* of the per-construction cross-sectional τ (Table 4): for each pair, the within-seed difference $\tau_k^{\text{pair-}B} - \tau_k^{\text{pair-}A}$, averaged over the 16 seeds with a Student-t CI, is the direct test of whether the two entry constructions change the ranking (not merely whether their between-seed intervals happen to overlap).

Multiplicity (M3): these six paired contrasts form one correlated family (they share the same 16 seeds and reuse per-carrier τ), so raw p-values are reported and the family-wide comparison is the pattern that all four $B2$-involving contrasts exceed the band while neither $B1$-A contrast does; a Bonferroni threshold of $\alpha/6 = 0.0083$ over the six contrasts leaves the four $B2$-involving results significant (smallest p among them 3×10^{-8}; the two $B1$-A p-values 0.009, 0.001 straddle it).

4.2 SAME-SAMPLE PAIRED IDENTIFICATION

The cross-sectional τ in Table 2 is not by itself decisive about path ordering: because $B2$ routes high-reference-influence samples early and that score correlates with realized value, a truly set-ordered world can still yield a large negative τ. We therefore measure each target's marginal value at multiple entry positions within the *same* seed, holding every other target at its frozen $B2$ step. For seed k and target z, the paired difference is $\Delta(z) = V(0.15T; z) - V(0.85T; z)$, by construction a *within-sample* contrast holding every other target at its frozen $B2$ step. The seed-level slope is

$$g_k = \frac{\text{mean}_{z \in G_k}\, \Delta(z)}{\text{std}_{z \in G_k}\, \Delta(z)}, \tag{4}$$

with the standard deviation across the 40 targets (ddof $= 1$); g_k is the standardized within-seed paired effect, so it is scale-free across seeds and carriers. **Sign convention:** $g_k < 0$ means early entry *lowers* a target's value (V at $0.15T$ below V at $0.85T$); $g_k > 0$ means early entry *raises* value. This is the *opposite* interpretation of the cross-sectional τ of Section 2: there, $\tau < 0$ means targets scheduled *earlier* are ranked *higher* in marginal value (a routing/rank statement), whereas here $g < 0$ means a target's value is *lower* when that same target is moved earlier (a causal-position statement). We therefore never equate the two signs; a negative τ and a negative g are *not* evidence of the "same direction," and we flag their reversal explicitly where both appear. The frozen preregistered decision uses a Student-t interval over the 16 seeds and a two-sided Wilcoxon signed-rank corroboration, with a minimal-effect band $m_{\text{eff}} = 0.10$. The result is carrier-dependent (Table 5).

On sine, the same-sample test returns a negative paired departure: the slope is negative, its interval excludes zero and lies beyond the minimal-effect band in the early-lowers-value direction. Because a negative g means early entry lowers value while a negative cross-sectional τ ranks early-scheduled targets higher (Section 2), the paired arm *reverses* the cross-sectional sign rather than reproducing it in the same direction (supplementary evidence for the undecided verdict). A per-seed sign-consistency fraction is not evidence: it is the no-test-power identity disclosed in Table 5. A routing-permutation placebo leaves the effect essentially unchanged ($g = -0.176$ vs. -0.186), rul-

Pair	linear	sine	composite
$B2$ vs A	**SEPARATED** $[-0.384, -0.257]$ vs $[-0.017, +0.032]$	**SEPARATED** $[-0.502, -0.440]$ vs $[-0.130, -0.075]$	separated on both
$B2$ vs $B1$	**SEPARATED** $[-0.384, -0.257]$ vs $[-0.125, -0.005]$	**SEPARATED** $[-0.502, -0.440]$ vs $[-0.056, +0.033]$	separated on both
A vs $B1$	overlap (amount 0.011)	**SEPARATED** $[-0.130, -0.075]$ vs $[-0.056, +0.033]$	mixed

Table 3: Frozen construction-axis CI separation. The registered rule assigns approximately zero construction-dependence weight if all pairs overlap on a carrier; separation of $B2$ from both A and $B1$ makes the $B2$-only threshold crossing construction-separable.

Same-seed pair	$\Delta\tau$	95% CI	p	$\lvert\Delta\tau\rvert$ beyond band
linear, $B2 - A$	-0.328	$[-0.395, -0.260]$	3×10^{-8}	yes
linear, $B2 - B1$	-0.256	$[-0.325, -0.186]$	1×10^{-6}	yes
linear, $B1 - A$	-0.072	$[-0.124, -0.021]$	0.009	no
sine, $B2 - A$	-0.369	$[-0.421, -0.317]$	2×10^{-10}	yes
sine, $B2 - B1$	-0.460	$[-0.509, -0.411]$	3×10^{-12}	yes
sine, $B1 - A$	$+0.091$	$[+0.041, +0.141]$	0.001	no

Table 4: Same-seed paired construction contrasts from frozen per-seed τ (16 seeds; no new computation beyond forming the within-seed differences). $\lvert\Delta\tau\rvert$ beyond band = the CI of the paired difference lies strictly outside $[-\tau_a, \tau_a] = [-0.10, +0.10]$. The $B2$ vs. A and $B2$ vs. $B1$ paired contrasts exceed the minimal-effect band on both carriers, while $B1$ vs. A does not—so $B2$'s region-lumped routing, not the per-sample position manipulation itself, is what separates $B2$ from the other two constructions.

Carrier	paired slope g	4-state label	routing-robust
sine	-0.186 CI$[-0.258, -0.115]$ Wilcoxon $p = 9 \times 10^{-5}$	**PATH-SUPPORTED-neg** (early entry lowers value)	placebo $g = -0.176$
linear	$+0.010$ CI$[-0.112, +0.132]$ Wilcoxon $p = 0.74$	**AMBIGUOUS** (not-identified)	—

Table 5: Same-sample paired identification (frozen R139/R140/R141 artifacts, 16 seeds). The sine paired test returns a negative g (interval excludes zero, beyond the minimal-effect band) that *reverses* the cross-sectional τ's sign interpretation (negative g: early entry lowers value; negative τ: early-scheduled targets rank higher; Section 4.2); it persists under routing permutation. The linear test is ambiguous. A per-seed sign-consistency fraction is deliberately omitted from this table: under this construction the per-seed sign of the frozen $B2$ cross-sectional reference τ is a structural constant, so a sign-agreement fraction is an identity equal to the fraction of negative seed-level slopes g_k—bookkeeping only, no test power, and not used as evidence in the abstract, contributions, or conclusion. The earlier "cross-sec $B2$ τ is a confirmed selection confound on both carriers" claim rested on a corrected sign-consistency code bug and is revoked.

ing out value-conditioned routing. On linear, the interval covers 0, so the cross-sectional negative τ is not identified as a same-sample path effect.

4.3 Set-ordered ridge counter-arm: paired-detector calibration

Because a routing-axis placebo cannot decide the *path-versus-set* axis, we run the identical paired estimator on the pure set-ordered ridge of Section 4.4, whose marginal value is order-invariant by construction (inverse-Hessian leave-one-out influence on a strictly convex quadratic).

The sealed pre-registration fixes the contract before the run: $g \approx 0$ with a 16-seed CI covering 0 was intended to count as silent, and any CI excluding 0 as the paired detector firing on a set-ordered world (Appendix B). Table 6 reports the outcome on the canonical target allocation together with the complete allocation sweep. The result is a quantified, conservative calibration:

- **The canonical allocation literally fires the sealed binary contract:** its CI excludes 0 ($g = +0.102$, CI$[+0.028, +0.176]$; Wilcoxon $p = 0.011$). We disclose this literally and do not soften it.
- **The bias is non-negative on the order-invariant substrate:** the point estimate is positive on all four target allocations ($+0.102$, $+0.192$, $+0.049$, $+0.145$), with the smallest CI lower bound -0.066. The sign is not outlier-driven: 14 of 16 seeds are positive on the canonical allocation, and dropping the most extreme seed of either sign leaves the positive sign intact.
- **The sealed contract's consequence clause is two-sided (sign-blind):** it maps "CI excludes 0" to "headline falsified", which is valid only if the bias shares the headline's (negative) sign. The measured bias is *positive*, opposite in sign to the negative sine headline (-0.186), so a positive bias cannot produce a negative departure; this is a specification erratum in the sealed consequence clause, not a rewrite of the frozen criterion.

Substrate / target allocation	paired g	95% CI	Wilcoxon p	CI covers 0
Set-ordered ridge, canonical $X_s[0{:}40]$	$+0.102$	$[+0.028, +0.176]$	0.011	no (fires)
Set-ordered ridge, $X_s[40{:}80]$	$+0.192$	$[+0.130, +0.254]$	0.00015	no (fires)
Set-ordered ridge, $X_s[80{:}120]$	$+0.049$	$[-0.066, +0.165]$	0.298	**yes (silent)**
Set-ordered ridge, random slice	$+0.145$	$[+0.065, +0.225]$	0.006	no (fires)
sine carrier (headline, Table 5)	-0.186	$[-0.258, -0.115]$	9×10^{-5}	no

Table 6: Set-ordered ridge counter-arm and its target-allocation sweep (16 seeds per row; canonical row from the sealed R148 artifact, sweep rows from the R148 finding record and code listed in Appendix B). *Takeaway:* the paired estimator's bias on the order-invariant world is *non-negative*—positive point estimates on all four target allocations ($+0.102/+0.192/+0.049/+0.145$, smallest CI lower bound -0.066)—so it cannot explain, and is conservative with respect to, the negative sine departure (-0.186). The canonical allocation literally fires the sealed binary contract (CI excludes 0); that contract's sign-blind consequence clause (specification erratum) does not falsify the negative headline. The sweep characterizes the reliability of the bias sign, not a rewrite of the frozen criterion.

Carrier	operating corr. ρ	routing-only τ	frozen $B2$ τ	zero-corr. τ
linear	0.30	-0.313	-0.321	-0.054
sine	0.55	-0.563	-0.471	-0.054

Table 7: B2 as a reference-conditioned stress test (frozen R141 artifact, 16 seeds). *Takeaway:* on a truly set-ordered ridge, value-conditioned routing at the real $B2$ operating correlations reproduces the full frozen $B2$ magnitudes, while at zero correlation the cross-sectional τ is near zero; the $B2$ cells are therefore reference-conditioned stress tests, not a standalone decisive path effect. Detector calibration is reported separately in Table 8.

Consequently the counter-arm does not falsify the *negative* sine departure; it quantifies a *non-negative* bias that a non-negative bias cannot explain (conservative: a cleaner zero baseline would make the sine departure only stronger). The post-hoc allocation sweep is used solely to establish the reliability of the bias sign, not to rewrite the already-fired frozen binary criterion. We deliberately do not report a magnitude-offset figure (such as -0.288) as a headline, because the offset's magnitude is not transferable from the ridge substrate to the sine carrier; only its sign is.

Framing choice (settled here). We do *not* adopt a "re-baseline" framing that subtracts the ridge offset magnitude from the sine reading, because that magnitude would need to transfer across carriers (ridge $\to$ sine, untested, unclaimed). We adopt the *sign-only, non-transport* conservative claim: a non-negative set-ordered bias cannot explain a *negative* sine departure. The cost is that we do not assert a quantitative *clean-null* (bias-corrected) calibration of the sine reading. A future re-baseline with magnitude transfer would require a *carrier-internal* order-invariant control (an order-invariant variant of the sine, linear-GD, WSD carrier from the same feature process), pre-registered here as a minimal 0-GPU follow-up rather than run, as it lies outside the frozen evidence.

4.4 B2 AS A REFERENCE-CONDITIONED STRESS TEST

On a set-ordered ridge, we route targets by a reference score with controlled Kendall correlation to realized value. At the real $B2$ correlations this reproduces the frozen magnitudes; at zero correlation it is near zero (Table 7). Thus $B2$ is a reference-conditioned stress test, not evidence that a path effect exists; the paired reading remains subject to Section 4.3. Decoupled positive and order-invariant negative controls provide the calibration rates in Table 8.

5 SAME-BED COMPARISON WITH A SET-VALUED BASELINE

On identical carriers, targets, held-out splits, and constructions, we compute per-seed Kendall agreement τ_{AB} between empirical path values and Equation 3. Table 9 separates $B1$ from canonical $B2$; all point estimates are positive, even with a negative paired departure on sine and an unidentified linear reading.

Frozen failure criterion. This arm is decisive only if (i) the bed is DISCRIMINATING, (ii) canonical $B2$ granularity is not mixed with A or $B1$, and (iii) agreement separates from chance. Otherwise it has approximately zero contribution weight. Because the frozen τ_{AB} estimates lack uncertainty intervals, Table 9 is descriptive.

Detector control	family	fires	rate
Decoupled positive control (imposed schedule-dependent influence, independent of the $B2$ routing score)	16 seeds	14	87.5%
Order-invariant negative control, unadjusted $\alpha = 0.05$	100 trials	3	3%
Order-invariant negative control, FWER-controlled frozen detector	20 seeded trials	0	0% (SILENT)

Table 8: Frozen detector calibration for the cross-sectional readout. *Takeaway:* the detector has genuine power on a decoupled substrate whose schedule dependence is imposed independently of the $B2$ routing score, and it stays at or below its nominal false-alarm level on an order-invariant substrate—silent under the frozen FWER-controlled setting also recorded in Table 10. The family size is stated for every rate so the unadjusted and FWER-controlled numbers are not read off the same trial count.

Carrier	$B1$ rank-linear	$B2$ region-lumped (canonical)
linear	$+0.20$	$+0.30$
sine	$+0.57$	$+0.55$

Table 9: Frozen mean within-construction rank agreement τ_{AB} between empirical path value and the inverse-Hessian set-valued comparator. Values are descriptive point estimates; the canonical same-bed arm is the $B2$ column.

6 Bed-discriminability certificate

The certificate combines a known path-sensitive probe, known set-ordered probe, and set-estimator aliveness check without treating $B2$ as path evidence. Table 10 records high power on the decoupled positive control, a silent negative control, and cross-sample estimator variation. Because routing reproduces the $B2$ firing on a set-ordered world, $B2$ cannot certify path sensitivity; the certificate validates both outcomes but does not replace the head-to-head uncertainty requirement.

Three probe properties should be kept separate, because a single statistic can have some without others: (i) *alive* (its readout varies across targets), (ii) *power for a known path effect* (it fires on a substrate with a manually imposed, schedule-dependent influence), and (iii) a *controlled false-alarm rate on a set-ordered substrate* (quiet when value is order-invariant). The $B2$ cross-sectional probe is alive and fires, but is only a *routing-sensitive stress probe*: its power for path-dependence is not established (reference-conditioned entry, firing reproduced by routing on a set-ordered world). The decoupled positive control is the probe with (ii); the NEG-NC order-invariant negative control is the probe with (iii). No single probe here is certified on all three, and the headline does not require it—the decisive evidence for entry-position sensitivity is the same-sample paired arm (Section 4.2), not the cross-sectional probes.

Reading	**linear**	**sine**
KNOWN-PATH probe: decoupled positive control (imposed schedule dependence, independent of $B2$)	✓ (14/16, 87.5%)	✓ (14/16, 87.5%)
$B2$ routing-sensitive stress probe (cross-sectional; NOT KNOWN-PATH)	✓ ($\tau = -0.321$)	✓ ($\tau = -0.471$)
KNOWN-SET: NEG-NC fires	0/20 SILENT	0/20 SILENT
Set estimator alive (frac_distinct)	1.0	1.0
Live τ_{AB} (certificate aliveness read)	$+0.88$	$+0.72$
Verdict	**DISCRIMINATING**	**DISCRIMINATING**

Table 10: Frozen bed-discriminability certificate. The known-path probe is the decoupled positive control (imposed schedule-dependent influence independent of the $B2$ routing score, $14/16$ detected); the $B2$ cross-sectional probe is listed as firing but *not* as KNOWN-PATH, because its signal is reproduced by value-conditioned routing on a set-ordered world (Section 4.4). The known-set probe remains silent, and the set-valued estimate has distinct values for every target. The aliveness read is part of the certificate and is distinct from the canonical head-to-head point estimates in Table 9.

7 RELATED WORK

Set-indexed data attribution. Datamodels predict behavior from training-set membership (Ilyas et al., 2022), while TRAK aggregates projected training gradients (Park et al., 2023). Both are indexed by selected *sets*, with no schedule-position axis. We test whether empirical leave-one-out value has a schedule-dependent component that such scores omit, varying entry rules while holding target identities and held-out readout fixed.

Uncertainty and reproducibility. Machine-learning conclusions can be sensitive to statistical power, run-to-run variance, and reporting choices (Card et al., 2020; Bouthillier et al., 2021; Dodge et al., 2019; Reimers & Gurevych, 2017). We therefore fix the seed as the inference unit, show confidence intervals rather than a single run, and keep a minimum-effect decision distinct from significance. Training-trajectory suites motivate retaining checkpoints and schedules as first-class experimental objects (Biderman et al., 2023); our contribution isolates one narrower object, sample entry construction. The frozen analysis plan follows the broader rationale for preregistration in empirical NLP and machine learning (van Miltenburg et al., 2021).

8 LIMITATIONS AND SCOPE

The evidence is restricted to deterministic CPU logistic-regression carriers with synthetic conflicts; the registered `sine` carrier does not represent general nonlinear models or large language models. The primary arm is the paired test; $B2$ is only a routing-reproducible stress test. The same-sample paired test's bias on the set-ordered ridge is *non-negative* ($g = +0.102$, CI$[+0.028, +0.176]$ on the canonical allocation, positive on all four allocations, Section 4.3); we report this as a quantified calibration and disclose that the canonical allocation literally fires the sealed binary contract, whose sign-blind consequence clause is a specification erratum that does not falsify the negative sine departure. Linear is not identified; the earlier selection-confound claim rested on a corrected sign-consistency bug. The decoupled control's 87.5% power is not evidence for canonical $B2$.

The comparator is an inverse-Hessian GLM reduction, not end-to-end TRAK/datamodel; its τ_{AB} values lack frozen uncertainty and are descriptive. One pre-registered set-ordered negative control cannot exclude every artifact or calibrate natural data. A natural carrier, full same-bed baselines, and agreement uncertainty remain necessary and unclaimed.

Future work: bed transfer to a larger carrier. Transfer the same three-entry-rule, same-seed, same-sample design to a natural-data or LLM-finetuning carrier, re-running the negative control, four-state margin readout, and bed certificate. The carrier-agnostic contract and $\tau_a = 0.10$ are pre-registered, preventing outcome tailoring. This tests nonconvex, diverse-data transfer; it does not replace same-bed baseline positioning or fresh review.

9 CONCLUSION

Whether per-sample marginal value is path-ordered or set-ordered is a *carrier-dependent measurement*, not a universal property, and this controlled, zero-GPU bed leaves the question honestly *undecided*. A frozen, pre-registered same-sample paired test controls the value-conditioned routing confound and returns carrier-dependent readings: the sine carrier shows a negative paired departure ($g = -0.186$, CI$[-0.258, -0.115]$) that *reverses* the cross-sectional sign (early entry lowers value; negative cross-sectional τ ranks early-scheduled targets higher), while the linear carrier is ambiguous ($g = +0.010$, CI$[-0.112, +0.132]$). A set-ordered ridge counter-arm quantifies the estimator's calibration: its bias is non-negative ($g = +0.102$, CI$[+0.028, +0.176]$ on the canonical allocation, positive on all four allocations), so it cannot explain—and is conservative with respect to—the negative sine departure. The canonical allocation literally fires the sealed binary contract, whose sign-blind consequence clause is a specification erratum that does not falsify the negative sine reading; we disclose the letter-firing and do not rewrite the frozen criterion. Its magnitude remains uncalibrated across carriers. Value-conditioned routing reproduces the full $B2$ magnitude on a set-ordered world, making $B2$ a stress test; set-ordering remains a coarse-graining, not a universally decidable property on this evidence.

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A Reproducibility and disclosure

All runs are deterministic CPU experiments with the seed as statistical unit. The quantitative figure is generated from a SHA-asserted frozen R128 JSON. Generative tooling assisted language/typesetting and the semantically checked qualitative schematic, which contains no measurements; a deterministic same-label glyph repair changed no module, arrow, or meaning. No generated asset created or modified quantitative evidence.

Ethics. The study uses synthetic data and no human participants, personal information, or user-generated content.

B Frozen provenance

The sixteen sealed specifications, outputs, derivations, and counter-arm artifacts are mapped below by SHA-256 prefix; full hashes and claim links remain in the author-side evidence ledger.

`61894856` R127 order run: `candidateB_order_r127.json`
`2e08b7f9` R126 reconcile: `candidateB_reconcile_r126.json`
`3a1c18da` R128 four-state labels: `candidateB_relabel_4state_R128.json`
`3178724b` R128 bed certificate: `candidateB_bed_discriminability_R128.json`
`40b74cbc` R133 construction CI: `candidateB_construction_axis_CI_R133.json`
`d74eb619` R127 supplement: `CANDIDATE_B_PREREG_SUPPLEMENT_R127_ORDERCONSTRUCTION.md`
`d0c898fb` R126 preregistration: `CANDIDATE_B_PREREG_v1_draft.md`
`547978c5` R139 same-sample paired and placebo: `candidateB_path_placebo_R139.json`
`b7028282` R140 rebuild (re-frozen R214): `candidateB_rebuild_R140.json`
`f6ba5097` R140 rebuild spec: `CANDIDATE_B_PREREG_SUPPLEMENT_R140_REBUILD.md`
`502c13bf` R141 preregistration (Rev. 2): `CANDIDATE_B_PREREG_SUPPLEMENT_R141.md`
`67c52c0b` R141 stress test and routing placebo: `candidateB_R141.json`
`e3431b7f` R148 set-ordered ridge counter-arm preregistration: `CANDIDATE_B_PREREG_SUPPLEMENT_R148_SETORDERED_RIDGE_ARMA.md`
`6b437ba5` R148 counter-arm output, canonical allocation: `candidateB_set_ordered_ridge_armA_R148.json`
`397dd466` R148 counter-arm code: `candidateB_set_ordered_ridge_armA.py`
`b81168c4` R148 counter-arm finding record, including the target-allocation sweep of Table 6: `R148_SETORDERED_RIDGE_ARMA_FINDING.md`

Full hashes and claim-level mappings are stored with the manuscript's auditable evidence ledger.